

Local Dissipative Decoder for 1D Repetition Code

Ising Hamiltonian Simple Approach

One simple approach is using a zero external field 1D Ising model Hamiltonian to define the energy of a given figuration of 'spins', which we are still dealing with as classical bits that may be 0 or 1.

$$H(\sigma) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j$$

We can calculate our set of syndromes, and I define the syndrome on each site to be given by

$$s_i = \sigma_i \sigma_{i+1} \text{ where } \sigma_i \in \{-1, 1\}$$

(Currently in my code I still have my state in terms of 0s and 1s and then I calculate the syndrome as, if neighboring sites are equal store 1 and if not store -1)

This allows me to rewrite the Hamiltonian in terms of syndromes as:

$$H(s) = - \sum_k s_k$$

However note that every site on the original state is involved in two syndromes, so the local energy at a site is given as follows, where I will from now on drop the constant $-J$

$$H_i = \sigma_{i-1} \sigma_i + \sigma_i \sigma_{i+1} \equiv s_{i-1} + s_i$$

We then propose a flip at a given site i and calculate the local change in energy that would result from this proposed flip. The the syndrome after the proposed flip is $s'_i = -s_i$, and so the local change in energy is given by

$$\Delta H_i = H'_i - H_i = 2J(s_{i-1} + s_i)$$

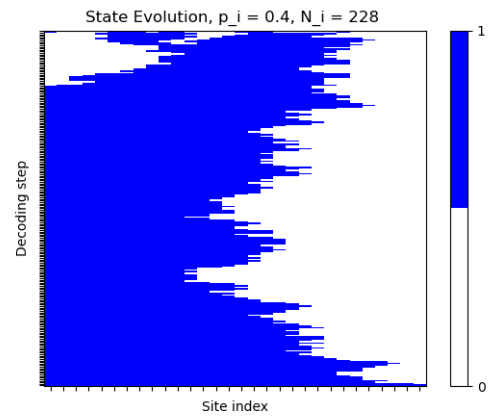
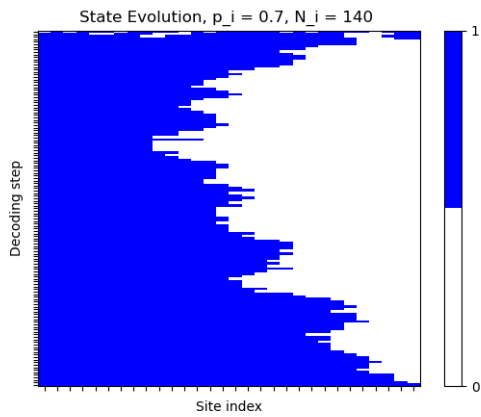
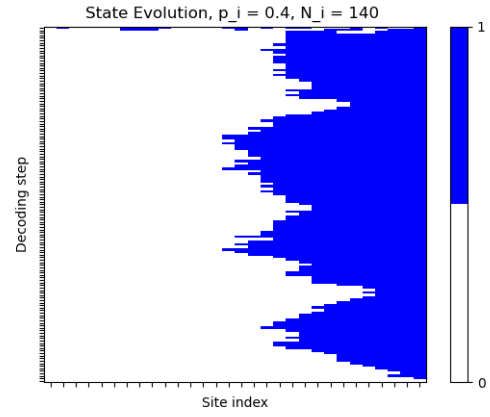
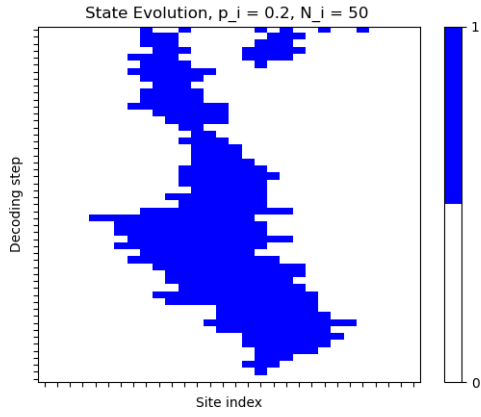
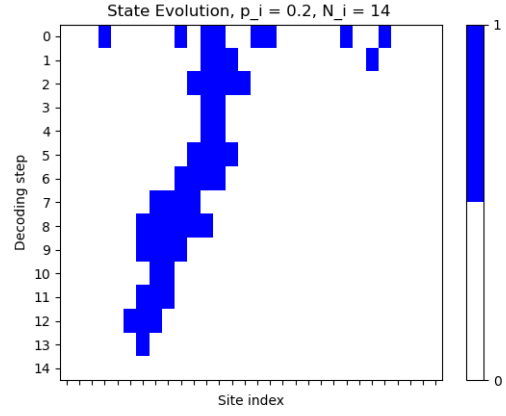
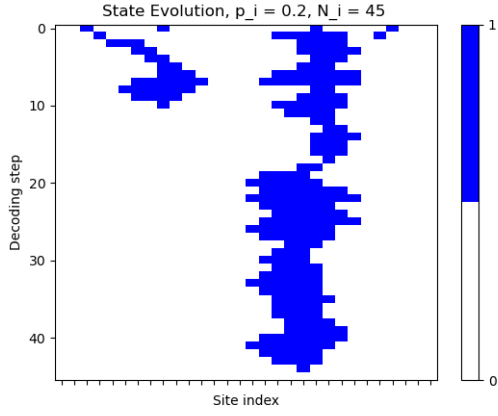
Our update rule is then:

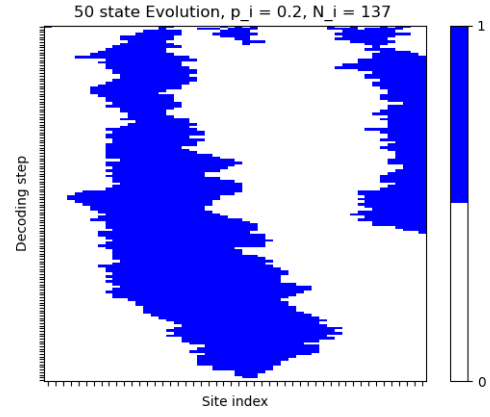
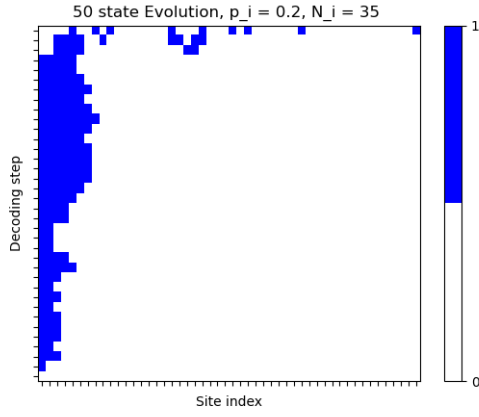
- if $\Delta H_i < 0 \rightarrow$ accept the flip because energy is decreased.
- if $\Delta H_i > 0 \rightarrow$ reject the flip because energy is increased.
- if $\Delta H_i = 0$ accept the flip with 1/2 probability.

I apply this update rule by choosing a random ordering for the program to perform a sweep of the entire state (maybe reconsider). Then I apply the update rule sequentially, meaning I apply the change to a given site, update my entire state and use that for the next suggested site alteration.

Some Results

What I index as one single decoding step in these graphs should actually be multiplied by N_i for the actual amount of steps computationally, as I update one site at a time and then sweep across all sites in random order.





1 Questions

- How to deal with degeneracy of the ground state? Is it no issue? Maybe add a small external field distribution to show preference over one ground state over the other.
- The way I implement the updates now is I calculate flip location across the entire state, then perform local energy analysis on each of these...
- when the decoder converges to the 1s ground state we consider it to have failed. It makes sense that at lower N it converges correctly less reliable because it reflects less accurately the p_i we are imposing on it (i.e. if I give it $p = 0.3$ in a very small initial state this might translate

2 TO DO

- Massively optimize code, you are calculating the syndromes inefficiently.
- Test scalability of the code as you increase the number of states, maybe average number of iterations needed to decode as you increase N . Or maybe a simpler metric like percentage runs that have a number of iterations over a certain threshold.

3 Results graveyard

Back when I was updating sites in parallel:

